

```
def rat_num_row():  
    f = open('vijay.txt', 'w+')  
    f.write("\nleft Angle Triangle Number Row\n-----\n")  
    for i in range(1,6):  
        for j in range(0,i):  
            f.write(str(i))  
        f.write("\n")  
    f.close()
```

rat_num_row()

```
def rat_num_col():  
    f = open('vijay.txt', 'a+')  
    f.write("\nleft Angle Triangle Number Row\n-----\n")  
    for i in range(1,6):  
        for j in range(i):  
            f.write(str(j))  
        f.write("\n")  
    f.close()
```

rat_num_col()

```
def ratrunner():
```

Ln: 1 Col: 0

```
def ratupper():  
    f = open('vijay.txt', 'a+')  
    f.write("\nleft Angle Triangle upper Row\n-----\n")  
    for i in range(1,6):  
        for j in range(6,i):  
            f.write(str(i))  
        for k in range(0,i):  
            f.write(chr(i+96))  
        f.write("\n")  
    f.close()
```

ratupper()

```
def ratlower():  
    f = open('vijay.txt', 'a+')  
    f.write("\nleft Angle Triangle upper column\n-----\n")  
    for i in range(1,6):  
        for j in range(6,i):  
            f.write(str(i))  
        for k in range(0,i):  
            f.write(chr(k+65))  
        f.write("\n")  
    f.close()
```

ratlower()

```
def rat_star():  
    f = open('vijay.txt', 'a+')  
    f.write("\nleft Angle Triangle star\n-----\n")  
    for i in range(1,7):  
        for j in range(i):  
            f.write(str("*"))  
        f.write("\n")  
    f.close()
```

rat_star()

```
def ratupper():  
    f = open('vijay.txt', 'a+')  
    f.write("\nleft Angle Triangle upper Row\n-----\n")  
    for i in range(1,8):  
        for j in range(8,i):  
            f.write(str(i))  
        for k in range(0,i):  
            f.write(chr(k+64))  
        f.write("\n")  
    f.close()
```

ratupper()



```
def ratlower():
    f = open('vijay.txt', 'a+')
    f.write("\nleft Angle Triangle lower column\n-----\n")
    for i in range(1,7):
        for j in range(7,i):
            f.write(str(i))
        for k in range(0,i):
            f.write(chr(k+97))
        f.write("\n")
    f.close()

ratlower()

def ratname_row():
    name = "vijay"
    f = open("vijay.txt", "a+")
    f.write("left Angle Triangle name lower row\n-----\n")
    for i in range(len(name)):
        for j in range(i+1):
            f.write(name[i])
        f.write("\n")
    f.close()

ratname_row()
```



```
def ratname_col():  
    name = "vijay"  
    f = open("vijay.txt", "a+")  
    f.write("left Angle Triangle name lower column\n-----\n")  
    for i in range(len(name)):  
        for j in range(i+1):  
            f.write(name[j])  
        f.write("\n")  
    f.close()
```

ratname_col()

```
def ratname_row():  
    name = "VIJAY".upper()  
    f = open("vijay.txt", "a+")  
    f.write("left Angle Triangle name upper row\n-----\n")  
    for i in range(len(name)):  
        for j in range(0, i+1):  
            f.write(name[j])  
        f.write("\n")  
    f.close()
```

ratname_col()

```
def ratname_col():  
    name = "VIJAY".upper()  
    f = open("vijay.txt", "a+")  
    f.write("left Angle Triangle name lower column\n-----\n")  
    for i in range(len(name)):  
        for j in range(0, i+1):  
            f.write(name[j])  
        f.write("\n")  
    f.close()  
  
ratname_col()  
  
def rat_num_inrow():  
    f = open("vijay.txt", "a+")  
    f.write("left Angle Triangle Number inverse row\n-----\n")  
    for i in range(7, -1, -1):  
        for j in range(0, i):  
            f.write(str(i))  
        f.write("\n")  
    f.close()  
  
rat_num_inrow()
```

```
def rat_num_incol():  
    f = open("vijay.txt", "a+")  
    f.write("left Angle Triangle Number inverse col\n-----\n")  
    for i in range(7, -1, -1):  
        for j in range(i):  
            f.write(str(j))  
        f.write("\n")  
    f.close()
```

```
rat_num_incol()
```

```
def rat_star_inverse():  
    f = open("vijay.txt", "a+")  
    f.write("left Angle Triangle star inverse \n-----\n")  
    for i in range(8, -1, -1):  
        for j in range(i):  
            f.write(str("*"))  
        f.write("\n")  
    f.close()
```

```
rat_star_inverse()
```

```
def rat_upper_incol():
```



```
def rat_upper_ir():  
    f = open('vijay.txt', 'a+')  
    f.write("\nleft Angle Triangle upper inverse Row\n-----\n")  
    for i in range(6,-1,-1):  
        for j in range(6,i):  
            f.write(str(i))  
        for k in range(0,i):  
            f.write(chr(i+64))  
        f.write("\n")  
    f.close()  
  
rat_upper_ir()  
  
def rat_upper_ic():  
    f = open('vijay.txt', 'a+')  
    f.write("\nleft Angle Triangle upper inverse column\n-----\n")  
    for i in range(6,-1,-1):  
        for j in range(6,i):  
            f.write(str(i))  
        for k in range(0,i):  
            f.write(chr(k+65))  
        f.write("\n")  
    f.close()  
  
rat_upper_ic()
```




```
def rat_lower_ir():
    f = open('vijay.txt', 'a+')
    f.write("\nleft Angle Triangle lower inverse Row\n-----\n")
    for i in range(8,-1,-1):
        for j in range(8,i):
            f.write(str(i))
        for k in range(0,i):
            f.write(chr(i+96))
        f.write("\n")
    f.close()

rat_lower_ir()

def rat_name_linr():
    name = "vijay"
    f = open("vijay.txt", "a+")
    f.write("left Angle Triangle name lower inverse row\n-----\n")
    for i in range(4,-1,-1):
        for j in range(5,i,-1):
            f.write(name[i])
        f.write("\n")
    f.close()

rat_name_linr()
```

```
def rat_name_linc():
    name = "vijay"
    f = open("viay.txt", "a+")
    f.write("left Angle Triangle name lower inverse column\n-----\n")
    for i in range(5, -1, -1):
        for j in range(4, i, -1):
            f.write(name[j])
        f.write("\n")
    f.close()
```

rat_name_linc()

```
def rat_name_uinr():
    name = "VIJAY".upper()
    f = open("vijay.txt", "a+")
    f.write("left Angle Triangle name upper inverse row\n-----\n")
    for i in range(4, -1, -1):
        for j in range(5, i, -1):
            f.write(name[i])
        f.write("\n")
    f.close()
```

rat_name_uinr()